Did you know?

The COVID-19 vaccine can alter your human DNA?

Did you know?

The COVID-19 vaccine has a microchip that tracks your location and movement?

Did you know?

That COVID-19 can spread through wireless and mobile networks using 5G?

Did you know?

It's All Fake News

Fake News Detection

ISM 6251

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Industry

- The primary application of this tool is combating misinformation by focusing on fake news detection.
- Fake news detection is critical in various industries, including:
    - Social Media
    - Journalism
    - Technology
    - Government
    - Healthcare
    - Etc.

Business Application

Social Platforms
Flagging tool to maintain user trust

Journalism
Screen press releases, news articles, and user-generated content

Tech Companies
Real-time news verification and data-driven decision-making

Government
Public remains accurately informed during crises and political campaigns

Healthcare
Monitor and counter health-related misinformation

And Many More!
The identification of fake news will benefit all industries

Motivation

- The primary motivation behind developing this fake news detection tool is to:
  - Address the growing threat of misinformation, which can undermine public safety, democracy, and organizational credibility.
  - Equip industries with innovative solutions to safeguard trust and maintain accurate public communication.
- Misinformation spreads rapidly and can damage reputations, mislead the public, and impact societal welfare. This machine learning-based tool provides:
  - A practical solution to detect and reduce fake news in real-time.
  - Support for industries like social media, journalism, government, and healthcare in protecting public trust and reducing the spread of misinformation.

Data Summary

Data

72,137 News Articles
35,028 Real
37,106 Fake

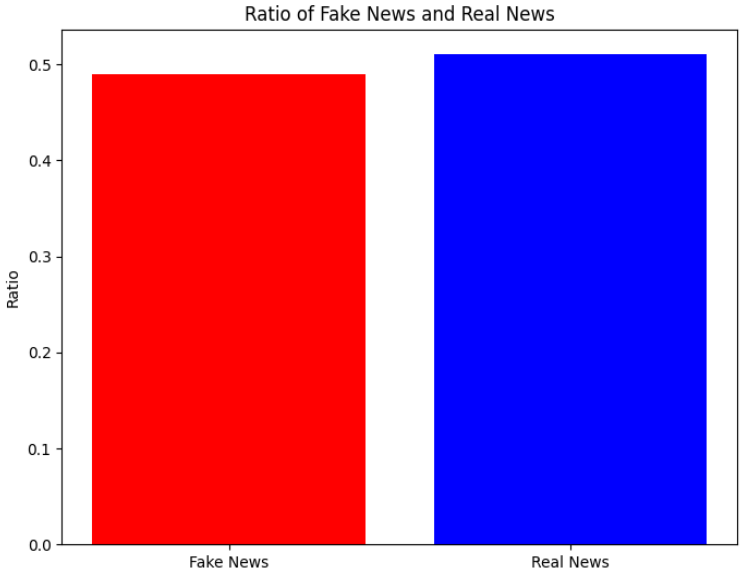
Features

Serial Number (starting from 0)
Title (about text news heading)
Text (about news content)

Dependent Variable

Label
0 = Fake
1 = Real

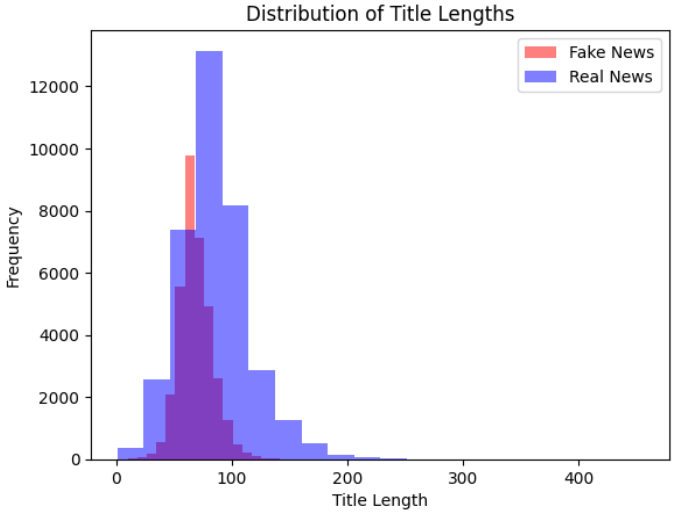
WELFake



Text & Title Length

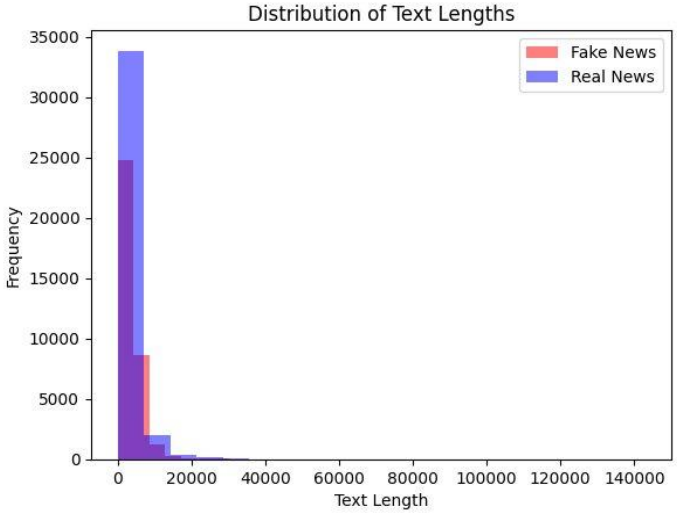
Title Length

Real News has a longer title at a much higher frequency than Fake News



Text Length

For both, frequencies drop significantly beyond 20,000 characters



Word Cloud

Real

New York Times

1

Trigram Word Cloud for Real News

Fake

President Donald Trump

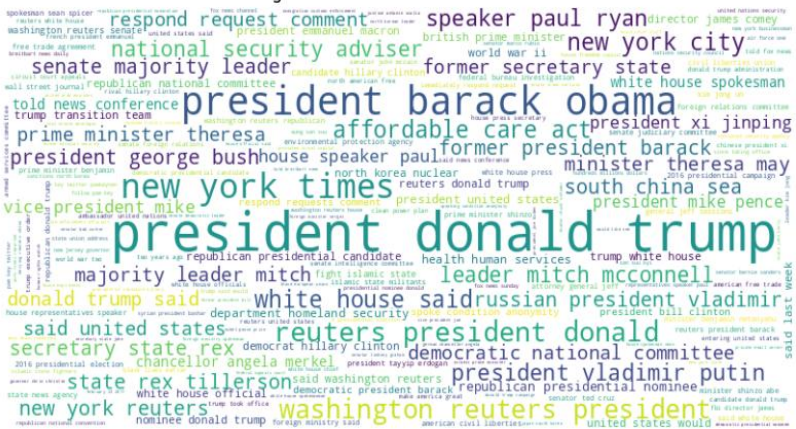
1

Real

Donald Trump Real

2

Trigram Word Cloud for Fake News



Fake

President Barack Obama

2

Model Overview

1

Goal: To build a real-time tool that detects fake news using machine learning algorithms

2

Algorithms Explored:

- Support Vector Machine (SVM)
- Logistic Regression
- Naive Bayes

3

Best Model: SVM

- Text Classification
- Precision
- Recall

Evaluation

1

Evaluation Criteria

Accuracy: The model's ability to correctly identify real vs. fake news articles.
Precision & Recall: Assessing the model's ability to avoid false positives and false negatives.
F1 Score: The balance between precision and recall.

2

Evaluation Results

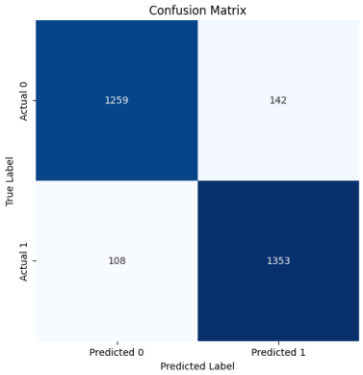
SVM Performance: The SVM model achieved the highest accuracy among the three models tested, making it the best-performing choice.
Precision: SVM effectively minimized false positives, ensuring that real news wasn't flagged incorrectly.
Recall: The model demonstrated strong recall, identifying most instances of fake news with few missed cases.
F1 Score: The SVM model balanced precision and recall well, providing a robust solution for fake news detection.

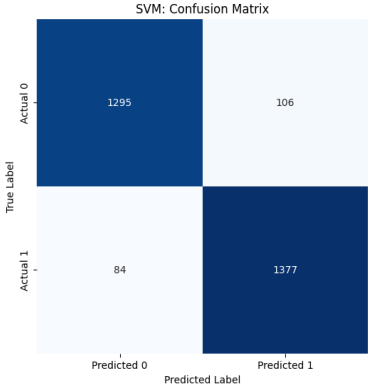
3

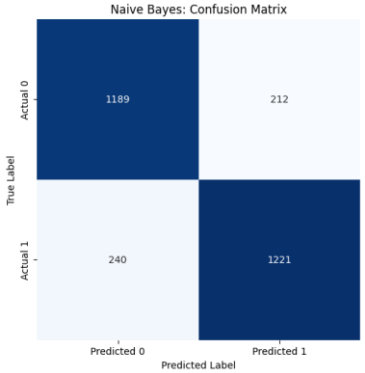
Confusion Matrix

Highlights the results of the confusion matrix, showing the performance of SVM in terms of true positives, false positives, true negatives, and false negatives.

Confusion Matrix







Logistic Regression

Type 1 Error: 142
Type 2 Error: 108

SVM

Type 1 Error: 106
Type 2 Error: 84

Naive Bayes

Type 1 Error: 212
Type 2 Error: 240

Business Value

Real-Time Detection:

Enables immediate identification and flagging of fake news to ensure credible information dissemination.

Impact on Public Trust:

By identifying fake news before it spreads, organizations can maintain credibility and reduce the reputational damage caused by misinformation.

Supports regulatory compliance and responsible digital behavior in platforms and businesses.

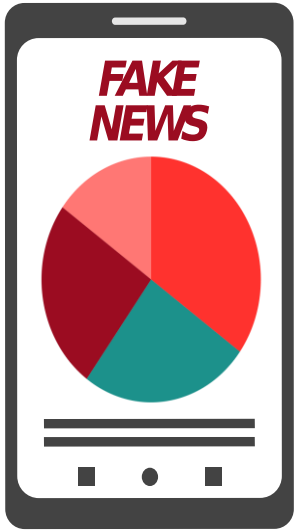
Industry Benefits:

Social Media: Prevents the spread of misinformation on platforms like Facebook and Twitter, preserving user trust.
Journalism: Helps media organizations maintain journalistic integrity by quickly verifying news sources.
Tech Companies: Provides AI-driven solutions for companies to integrate into their cloud-based services, enhancing data-driven decision-making.
Governments & Healthcare: Ensures the public receives accurate information during crises (e.g., elections, public health emergencies).

Long-Term Value:

Scalability: The model can be easily adapted and scaled to accommodate new forms of digital content and more complex data types.
Innovation in AI & ML: Promotes responsible AI use, positioning companies as leaders in the fight against misinformation.

Group Member Task Breakdown



Ryan

Report
Presentation Slides

Niharika

Report

Zahid

Python Workbook

Sucharitha

Report

Kai

Presentation Slides

References

- Mayo Clinic Staff. "COVID-19 Myths Debunked." *Mayo Clinic*, Mayo Foundation for Medical Education and Research, 3 Mar. 2023, www.mayoclinic.org/diseases-conditions/coronavirus/in-depth/coronavirus-myths/art-20485720.
- Martin, Emily. "Fake News, Social Media, and the Death of Truth." *CBS News*, CBS Interactive, 2 May 2023, www.cbsnews.com/news/fake-news-social-media-and-the-death-of-truth/.
- "Commentary: Tech Companies' Inability to Control Fake News Exacerbates Violent Acts." *Business & Human Rights Resource Centre*, Business & Human Rights Resource Centre, 18 Oct. 2023, www.business-humanrights.org/en/latest-news/commentary-tech-companies-inability-to-control-fake-news-exacerbates-violent-acts/.
- Columbia School of Professional Studies. "The Real Impact of Fake News: The Rise of Political Misinformation and How We Can Combat Its Influence." *Columbia University*, School of Professional Studies, 10 Sept. 2023, sps.columbia.edu/news/real-impact-fake-news-rise-political-misinformation-and-how-we-can-combat-its-influence.
- Office of the U.S. Surgeon General. "Confronting Health Misinformation." *U.S. Department of Health and Human Services*, 15 Aug. 2023, www.hhs.gov/surgeongeneral/priorities/health-misinformation/index.html.

Thank You